

Connor Sweet

Robotics and Full-Stack Software Developer

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Work Experience

Full-Stack Software Developer

ParkCo

February 2026 – Present

- Built a Node.js integration with third-party parking enforcement systems using SOAP/XML workflows to synchronize transaction data across distributed services
- Developed a Python proof-of-concept for a Raspberry Pi-based edge controller that abstracted hardware interfaces, prioritized real-time operational tasks through queue scheduling, and improved reliability through explicit error modeling
- Supported cloud deployments and production infrastructure using Docker and AWS (ECS, ECR, CloudWatch), including service rollbacks, environment configuration, and release troubleshooting
- Improved backend reliability by implementing API fixes across payment, permissions, and bulk-upload workflows, including redesigning error handling to replace unhandled failures with structured responses

Robotics Software Developer

Lincoln Electric Automation

July 2023 – January 2026

September 2022 – December 2022 (Co-op)

January 2022 – April 2022 (Co-op)

- Used Vue, TypeScript, fp-ts, Effect and Babylon.js to build and support a pendant-based robotic welding interface
- Developed a system to author user-taught robot actions as recognizable welding instructions for Fanuc and ABB robots
- Wrote an algorithm to convert the location and orientation of cartesian 3D points into relative frame representations, supporting coordinated motion between a 7 axis robot and an external rotary trunnion
- Created a multi-pass welding solution, leveraging quaternion-based transformations to interpret user-defined torch offsets
- Produced a method to support remote cycle execution of welding programs across robots using an operator panel
- Authored a data migration workflow for persistence of programmed weld parameters into revised format to enable welding with crater fill

Developer and Automation Specialist

Mach7 Technologies

January 2021 – April 2021 (Co-op)

- Created Dart API allowing PostMessage requests through commands sent by external applications
- Developed Javascript API to aggregate performance metrics from onscreen video in a clinical viewer
- Built infrastructure for frame rate performance testing in Java
- Implemented spine label DICOM markup manipulation through console commands in clinical viewer

Skills

Languages

TypeScript JavaScript Python C
C++ Java

Frameworks & Technologies

Node.js Express Vue.js React
Docker AWS MongoDB
PostgreSQL Raspberry Pi
Babylon.js

Education

BASc: Honours Computer Engineering - Artificial Intelligence Option

University of Waterloo

2017 – 2023

- Graduated with Distinction
- Elected Class Academic Representative
- Member of Waterloo Mars Rover and Robotics teams

Projects

Shapley Routing Python API

- Designed and integrated an algorithm to compute exact Shapley values for ride-sharing games, facilitating fair cost allocation among participants
- Developed an efficient approximation algorithm for estimating Shapley values in ride-sharing scenarios with many participants

Indoor GPS Navigation System

- Implemented building mapping workflow in React Native, allowing users to graph points of interest based on uploaded floor plans and GPS coordinates
- Developed a backend server in Django connected to a PostgreSQL database to store maps of multiple buildings with several floors